

Closed-Form Storage Stability, Induction Delay, and Thermal-Feedback Criticality of Melt-Gated Autocatalytic Cure

Leon Sandler

Independent researcher, Northbrook, Illinois, USA

Corresponding author: sandler.leon@gmail.com

ORCID: 0009-0007-4584-808X

Abstract

A cure triggered by the melting of an encapsulated catalyst combines a distribution of capsule melting temperatures, autocatalytic kinetics, and exothermic feedback. We derive what can be written in closed form for the lumped model and test each result against the full equations. (i) At constant temperature the gated cure is exactly the isothermal cure on the integrated availability of the catalyst; the delay of a fully activated population is $\tau_{\text{rel}}[1 - \exp(-t/\tau_{\text{rel}})]$, approaching the release time constant τ_{rel} only at late conversion. (ii) The isothermal cure time is exact by one quadrature. (iii) For Gaussian capsule melting temperatures of spread σ the storage conversion is $(a/b)[\exp(b f t_{\text{eff}}) - 1]$ with open fraction $f = \Phi(-\Delta/\sigma_{\text{eff}})$, which bounds σ to within 3 % of the population equations. (iv) The frozen-conversion thermal balance has an algebraic branch only for $\Pi = \psi\Theta$ below $\Pi_{\text{c}} = (1+n)^{(1+n)}/[e n^n (1+\epsilon)^{(1+n)}]$ (1.96 for $n = 3/2$); integrated thresholds approach it as $\Theta \rightarrow 0$ and, with Arrhenius kinetics, exceed it by about $\exp(1/\text{Ar})$ (within 5 % on the tested grid). (v) In a continuous-flow reactor the steady-state curve has two folds above $\kappa_{\text{c}} = 1.89$, but the upper branch can lose stability at a Hopf-type point before its fold; the window of two stable states and its loop area are computed, and finite-rate loops approach it slowly. All parameters are illustrative; no experimental data are used.

Keywords: autocatalytic cure kinetics; thermal runaway; encapsulated catalyst; Lambert W function; saddle-node and Hopf bifurcation; hysteresis

I. Introduction

Encapsulating a reactive component and releasing it on demand is a general way to separate the stability of a formulation from the speed of its cure. The best-known instance is the self-healing polymer, in which embedded capsules release a healing agent when damaged [1]; thermally triggered variants use the melting of a wax or polymer shell, whose phase-change behavior is the subject of a literature on microencapsulated phase-change materials [2] and whose release is described by a family of release models [3]. When the released species catalyzes a thermoset cure, three features interact: the capsule

population has a distribution of melting temperatures rather than a single one, the cure is autocatalytic and follows the form introduced by Kamal and Sourour [4, 5], and the cure heat raises the temperature of the specimen, which accelerates the cure. The cure exotherm makes the temperature field of thick thermoset sections depend on the cure history [6], and the feedback between heat release and rate is the subject of the classical theory of thermal explosion of Semenov and Frank-Kamenetskii [7, 8]. That theory has been extended to reactant consumption [9] and to the induction period [10], and the multiplicity and stability of the steady states of a heated, continuously fed reactor were analyzed by van Heerden, by Aris and Amundson, and by Uppal, Ray and Poore [11–14].

Calorimetric kinetic analysis is well codified [15, 16], and the separable form of the cure law is a standard route to closed-form cure times, with the temperature history entering through a generalized time [17]. In that literature the rate law and the thermal runaway of a specimen are usually treated separately. The question addressed here is narrower: for the lumped model of a *melt-gated* autocatalytic cure, which statements linking the quantities that a calorimeter measures (the rate constants of the cure, the width of the melting endotherm of the capsules, the thermal time constant of a specimen) to storage stability, delay and runaway can be written in closed form, and how well do they hold against the full equations?

The paper separates standard identities from results specific to this system. Proposition 1 is the generalized-time identity [17] applied to the catalyst availability; its content is the identification of the clock and the resulting delay law (Section III A). Proposition 2 collects standard integrals and one sharpness law (III B). Proposition 3, the storage conversion of a Gaussian capsule population and the bound on the melting-temperature spread that follows from it, is specific to the gated system (III C). Proposition 4 applies the Semenov fold of the frozen-conversion balance to the Kamal–Sourour consumption law, which gives a closed-form critical number, its overshoot below the fold and an Arrhenius correction (III D); the classical treatment of consumption [9] is not specific to autocatalysis. Proposition 5 gives the steady states, the Jacobian and the stable window of a continuous-flow reactor with this kinetics (III E). A synthetic-data test shows how the closed forms behave as fitting functions (III F). To the author's knowledge, in the sources examined, these results have not been assembled for a gated autocatalytic cure.

The illustrative parameter set and the capsule-population picture were introduced in a numerical study by the author [18]: the rate constants and orders of Table 1, the release rate constant, the logistic activation and the mean melting temperature, the adiabatic rise, and the reference integrator come from that study. The analytic results below are new, and the comparisons of that study with first-order controls are not used here. The model is lumped and every parameter is illustrative: nothing is fitted to measurements, and the paper makes no claim about a particular material.

II. Model and scaling

A. Equations

The conversion α of the resin obeys a Kamal–Sourour law with reaction order n for the unreacted fraction and first order in the autocatalytic term,

$$\frac{d\alpha}{dt} = \langle C \rangle(t) (k_1 + k_2 \alpha) (1 - \alpha)^n, \quad (1)$$

where $\langle C \rangle$ is the availability of the catalyst averaged over the capsule population (the fraction of the catalyst that has been released, $0 \leq \langle C \rangle \leq 1$). The temperature of the well-mixed specimen obeys a lumped energy balance with Newtonian loss and the adiabatic temperature rise ΔT_{ad} of the reaction,

$$\frac{dT}{dt} = \frac{(T_\infty - T)}{\tau_{\text{th}}} + \Delta T_{\text{ad}} \frac{d\alpha}{dt}. \quad (2)$$

Capsule i has a melting temperature $T_{m,i}$, drawn from a Gaussian of mean T_m and standard deviation σ , and releases its catalyst by first-order relaxation toward a logistic activation of width w_0 ,

$$\frac{dC_i}{dt} = k_{\text{rel}}(T) [G_i(T) - C_i], \quad G_i = \left(1 + \exp \left(- \frac{T - T_{m,i}}{w_0} \right) \right)^{-1}, \quad \langle C \rangle = \frac{1}{K} \sum C_i. \quad (3)$$

The initial state is $\alpha(0) = 0$ and $C_i(0) = 0$ unless stated, and $T(0) = T_\infty$ for the isothermal-start problems of Sections III A–D. The rate constants are Arrhenius, $k_j = A_j \exp(-E_j/RT)$, for $j = 1$ (background), 2 (autocatalytic) and rel (release). The model has no spatial structure, no diffusion control and no vitrification (Section VI). Throughout, the *illustrative* parameter set of Table 1 is used.

Table 1 Illustrative parameter set (not fitted to any material; taken from Ref. [18]). Arrhenius forms are $k = A \exp(-E/RT)$. Note that $E_1 \neq E_2$.

Quantity	Symbol	Value
Background rate constant	A_1, E_1	$2.0 \times 10^3 \text{ min}^{-1}$, 50 kJ mol ⁻¹
Autocatalytic rate constant	A_2, E_2	$2.5 \times 10^7 \text{ min}^{-1}$, 65 kJ mol ⁻¹
Reaction order	n	1.5 (1 and 2 where stated)
Release rate constant	$A_{\text{rel}}, E_{\text{rel}}$	$6.4 \times 10^6 \text{ min}^{-1}$, 60 kJ mol ⁻¹
Mean melting temperature, activation width	T_m, w_0	118 °C, 0.5 °C
Adiabatic rise, thermal time constant	$\Delta T_{\text{ad}}, \tau_{\text{th}}$	220 K, varied (Section III D)
Rate constants at 140 °C	k_1, k_2, k_{rel}	9.53e-04, 0.151, 0.166 min ⁻¹

B. Dimensionless groups and the exponential comparison model

Define $\varepsilon = k_1/k_2$, the thermal number $\Theta = \tau_{th}k_2(T_\infty)$, the reduced exothermicity $\psi = \Delta T_{ad}E_2/(RT_\infty^2)$, the activation number $Ar = E_2/(RT_\infty)$ and the release number $\Lambda = k_2/k_{rel}$. The combination that controls thermal feedback is

$$\Pi = \psi \Theta = \frac{\Delta T_{ad} E_2 \tau_{th} k_2}{R T_\infty^2}. \quad (4)$$

The *exponential comparison model* used in Sections III D and III E replaces the Arrhenius dependence of both rate constants by the common Frank-Kamenetskii form $k_i(T) = k_i(T_\infty) \exp(\psi\theta)$, $\theta = (T - T_\infty)/\Delta T_{ad}$, and treats ε as a constant. This is exact when the activation energies are equal and is a good description of the rate near the fold when the background term is negligible there: the fold occurs at conversions of order 0.4, where the autocatalytic term ($\alpha \approx 0.4$) exceeds the background term ($\varepsilon = 3 \times 10^{-3}$) by two orders of magnitude, so the different activation energy of k_1 (50 against 65 kJ mol⁻¹) matters little. The *Arrhenius simulations* of Sections III D (Table 5) do not use this simplification: both $k_1(T)$ and $k_2(T)$ follow their own Arrhenius law and $\varepsilon = k_1/k_2$ is evaluated at T_∞ .

C. Computation and use of AI assistance

All numerical results are produced by the released scripts (run_all.py writes results.json; every number in the text is read from it, none is typed by hand), with the reference integrator, solver settings and protocols given in Appendix A. The model code, the closed-form module, the tests, the figure scripts and a first draft of the text were produced with the assistance of Claude Sonnet 5.5 (model identifier claude-sonnet-5-5; Anthropic), used through the Claude Code command-line environment on the author's computer, because the author is an independent researcher without a computational group. The assistant wrote and ran code, drafted derivations and checked algebra; closed forms were verified against independent numerical integration by the test suite (code/test_theory.py) and by the comparisons reported in Section III, and every journal reference was resolved through Crossref. The assistant did not enter or alter any numerical result by hand. The author is responsible for the content.

III. Results

A. Proposition 1: the gated cure runs on the clock of integrated availability

Proposition 1. At constant temperature, with $\alpha(0) = 0$ and $C_i(0) = 0$, let $F(\alpha) = (k_1 + k_2\alpha)(1 - \alpha)^n$. Then the gated conversion obeys

$$\int_0^\alpha \frac{dx}{F(x)} = s(t), \quad s(t) = \int_0^t \langle C \rangle(t') dt'. \quad (5)$$

Proof. With T constant, $d\alpha/dt = \langle C \rangle(t)F(\alpha)$ is separable: $d\alpha/F(\alpha) = \langle C \rangle dt$. Integration from the initial state gives the identity. ■ This is the generalized-time identity [17] with the integrated availability as the clock. The left side is the isothermal time $t_{\text{iso}}(\alpha)$ of the ungated resin, so the gated time to reach α is the time at which $s(t) = t_{\text{iso}}(\alpha)$. For a population that is fully open at the operating temperature, $\langle C \rangle = 1 - \exp(-t/\tau_{\text{rel}})$ with $\tau_{\text{rel}} = 1/k_{\text{rel}}$, so $s(t) = t - \tau_{\text{rel}}[1 - \exp(-t/\tau_{\text{rel}})]$ and

$$t_{\text{gated}}(\alpha) - t_{\text{iso}}(\alpha) = \tau_{\text{rel}} \left(1 - \exp \left(- \frac{t_{\text{gated}}(\alpha)}{\tau_{\text{rel}}} \right) \right). \quad (6)$$

The delay is therefore conversion dependent: it lies between 0 and τ_{rel} and approaches τ_{rel} only when $t_{\text{gated}} \gg \tau_{\text{rel}}$. The fractional deficit from τ_{rel} is below 1 % when $t_{\text{gated}}/\tau_{\text{rel}}$ exceeds $\ln 100 = 4.605$. At 140 °C ($\tau_{\text{rel}} = 6.02$ min) the delay is 2.81 min at $\alpha = 0.001$ ($t_{\text{gated}}/\tau_{\text{rel}} = 0.63$), 5.12 min at $\alpha = 0.01$, and 5.94, 6.02 and 6.02 min at $\alpha = 0.1, 0.5$ and 0.9 (Table 2); the constant shift is a good description for $\alpha \gtrsim 0.1$ at this temperature and fails at early conversion. Independent adaptive integration of the gate and cure equations reproduces Eq. (6) to the integration tolerance in every row, and integrating the full model with the reference RK4 integrator at 125, 140 and 155 °C and $\alpha = 0.1, 0.5, 0.9$ reproduces the gated times of Eq. (5) to a relative error of at most $7.7\text{e-}07$.

For incomplete activation, with open fraction $f < 1$ (a fraction of the capsules that has not melted at the operating temperature), the clock is $s(t) = f[t - \tau_{\text{rel}}(1 - \exp(-t/\tau_{\text{rel}}))]$, so the delay relative to the ungated resin is not a constant shift: the gated cure is slowed in proportion to $1/f$ as well. For $f = 0.5$ at 140 °C the integrated times to $\alpha = 0.1, 0.5$ and 0.9 are 45.1, 78.5 and 138.9 min, and the clock gives 45.1, 78.5 and 138.9 min, while the fully open population reaches them at 25.5, 42.3 and 72.4 min. A population whose mean melting temperature lies below the operating temperature is not necessarily fully open: a broad population has a tail of capsules that melt above it, and f must be computed from the melting-temperature distribution (Section III C). The statement is exact only at constant temperature; a heat-up transient or an exotherm adds a temperature-dependent delay that Proposition 1 does not describe.

Table 2 Delay of a fully open population relative to the ungated resin at 140 °C ($n = 3/2$, $\tau_{\text{rel}} = 6.02$ min): integrated gate and cure equations against Eq. (6).

α	$t_{\text{iso}}(\alpha)$ (min)	t_{gated} (min), integrated	Delay (min)	$\tau_{\text{rel}}[1 - \exp(-t_{\text{gated}}/\tau_{\text{rel}})]$ (min)	$t_{\text{gated}}/\tau_{\text{rel}}$
0.001	0.97	3.78	2.81	2.81	0.63
0.01	6.32	11.45	5.12	5.12	1.90
0.1	19.55	25.49	5.94	5.94	4.23
0.5	36.24	42.26	6.02	6.02	7.01
0.9	66.42	72.45	6.02	6.02	12.03

B. Proposition 2: isothermal cure time, induction, and sharpness

Proposition 2. With $a = k_1$ and $b = k_2$, the isothermal time to reach conversion α is, for any n , $t_{\text{iso}}(\alpha) = \int_0^\alpha \frac{dx}{[(a + bx)(1 - x)^n]}$, and for $n = 1$

$$t_{\text{iso}}(\alpha) = \frac{1}{a + b} \ln \frac{a + b\alpha}{a(1 - \alpha)}. \quad (7)$$

Proof. Partial fractions of $1/[(a + bx)(1 - x)]$ give $(a + b)^{-1}[b/(a + bx) + 1/(1 - x)]$, and integration gives the logarithm. ■ For small α the factor $(1 - x)^n$ is close to 1, so the induction time ($\alpha = 0.1$) is almost independent of n ; the late-time tail is not. For $n = 1$ the sharpness $S = (t_{90} - t_{10})/t_{10}$ is exactly

$$S = \frac{\ln \left(\frac{9(a + 0.9b)}{a + 0.1b} \right)}{\ln \left(\frac{a + 0.1b}{0.9a} \right)} \quad (8)$$

and, for $b \gg 9a$ and $n = 1$ only, it reduces to the asymptotic law

$$S \approx \frac{2 \ln 9}{\ln \left(\frac{b}{9a} \right)}. \quad (9)$$

Sharpness is then controlled by the rate-constant ratio alone, and only logarithmically. At 140 °C, $b/a = 159$ gives $S = 1.48$ exactly for $n = 1$ and 1.53 from Eq. (9), with a difference of 3.4 %; the law is accurate to 5 % across 110–155 °C for $n = 1$. A hundredfold increase of b/a , from 159 to 15,900, lowers the exact S from 1.48 to 0.59, a factor of 2.5. For $n \neq 1$ the law does not apply: at 140 °C the integrated sharpness rises from 1.48 ($n = 1$) to 2.40 ($n = 3/2$) because of the longer tail, and the quadrature should be used. The closed forms and the quadrature reproduce the integrated t_{10} and t_{90} for $T = 110$ –155 °C and $n = 1$ and $3/2$ to within 1.2e-07 (relative).

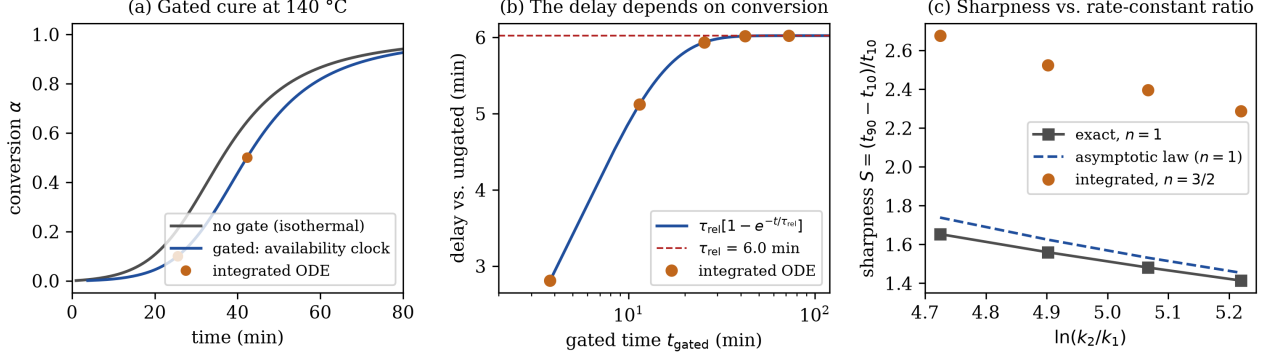


Fig. 1 (a) Gated and ungated cure at constant temperature (140 °C): the gated curve is the isothermal curve evaluated on the integrated-availability clock, and the integrated ODE (circles) agrees. (b) Delay of the gated cure relative to the ungated resin as a function of the conversion level: integrated (points), Eq. (6) (line), and the limit τ_{rel} (dashed). (c) Sharpness against the rate-constant ratio: exact $n = 1$, the asymptotic law Eq. (9), and integrated results for $n = 3/2$.

C. Proposition 3: storage conversion of a capsule population and a bound on the spread

A capsule that is open at the storage temperature T_s releases catalyst that drives the cure at T_s . For a Gaussian spread σ of melting temperatures and a logistic activation of width w_0 , the equilibrium open fraction at a margin $\Delta = T_m - T_s$ is the Gaussian–logistic convolution, which the probit approximation (matching the variance $\pi^2 w_0^2/3$ of the logistic) writes as

$$f = \Phi\left(-\frac{\Delta}{\sigma_{\text{eff}}}\right), \quad \sigma_{\text{eff}}^2 = \sigma^2 + \frac{\pi^2 w_0^2}{3}. \quad (10)$$

The approximation, not the identity, carries the error of this step; it is tested below. **Proposition 3.** If every capsule relaxes toward its own activation with the common rate constant $k_{\text{rel}}(T_s)$, then $\langle C \rangle(t) = f[1 - \exp(-k_{\text{rel}}t)]$ and, in the small-conversion limit ($\alpha \ll 1$, $(1 - \alpha)^n \approx 1$), the storage conversion after a time t is

$$\alpha_s(t) = \frac{a}{b}[\exp(bf t_{\text{eff}}) - 1], \quad t_{\text{eff}} = t - \tau_{\text{rel}}\left(1 - \exp\left(-\frac{t}{\tau_{\text{rel}}}\right)\right). \quad (11)$$

Proof. In the small-conversion limit the rate law is linear, $d\alpha/dt = \langle C \rangle(a + b\alpha)$, so that $\alpha = (a/b)[\exp(b s(t)) - 1]$ with $s(t) = \int \langle C \rangle dt = f t_{\text{eff}}$. ■ The tolerance η on the storage conversion (a symbol distinct from $\varepsilon = k_1/k_2$) then fixes the largest admissible open fraction exactly, $f^* = \ln(1 + \eta b/a)/(b t_{\text{eff}})$, and the bound on the spread follows by inverting Eq. (10), including w_0 :

$$\sigma^* = \sqrt{\left(\frac{\Delta}{z^*}\right)^2 - \frac{\pi^2 w_0^2}{3}}, \quad z^* = -\Phi^{-1}(f^*). \quad (12)$$

This is defined for $0 < f^* < 1/2$ (a finite positive bound). If $f^* \geq 1/2$ no finite upper bound on the spread is imposed, because even a population centered at the margin Δ above the storage temperature cannot

exceed the tolerance; if the radicand is negative, even $\sigma = 0$ violates the tolerance because the logistic width alone opens too large a fraction. Eq. (12) with $w_0 = 0.5$ °C generated Table 3; the column for $w_0 \rightarrow 0$ shows the limit, which differs from it most for the narrowest widths. We tested the closed forms against the capsule-population equations ($K = 801$ deterministic Gaussian quantiles of the melting temperature, constant temperature, $t = 150$ min) for margins of 3, 6 and 10 °C and spreads of 2–10 °C: the storage conversions agree within 4 % (difference = closed form/numerical – 1; median ratio 0.992), the closed form agrees with the defining integral to 4e-16, and the spread σ^* that solves $\alpha_s = \eta$ agrees with the root of the population equations within 3 % for all 8 cases in which the ODE reaches η (Table 3). For $\eta = 0.01$ and a 6 °C margin the bound is $\sigma^* = 5.9$ °C.

Table 3 Largest admissible melting-temperature spread σ^* for a storage time of 150 min and $w_0 = 0.5$ °C: Eq. (12), its $w_0 \rightarrow 0$ limit, and root-finding on the population equations (difference = closed form/ODE – 1).

Margin Δ (°C)	Tolerance η	f'	σ^* Eq. (12) (°C)	σ^* , $w_0 \rightarrow 0$ (°C)	σ^* population ODE (°C)	Difference
3	0.001	0.019	1.13	1.45	1.12	+1.4 %
3	0.01	0.134	2.55	2.71	2.52	+1.1 %
3	0.05	0.331	6.79	6.85	6.59	+3.0 %
6	0.001	0.022	2.84	2.99	2.84	+0.1 %
6	0.01	0.156	5.87	5.94	5.85	+0.4 %
6	0.05	0.389	21.34	21.36	21.90	-2.5 %
10	0.001	0.027	5.12	5.20	5.12	+0.0 %
10	0.01	0.194	11.53	11.57	11.53	+0.0 %
10	0.05	0.489	> 40	> 40	> 40	–

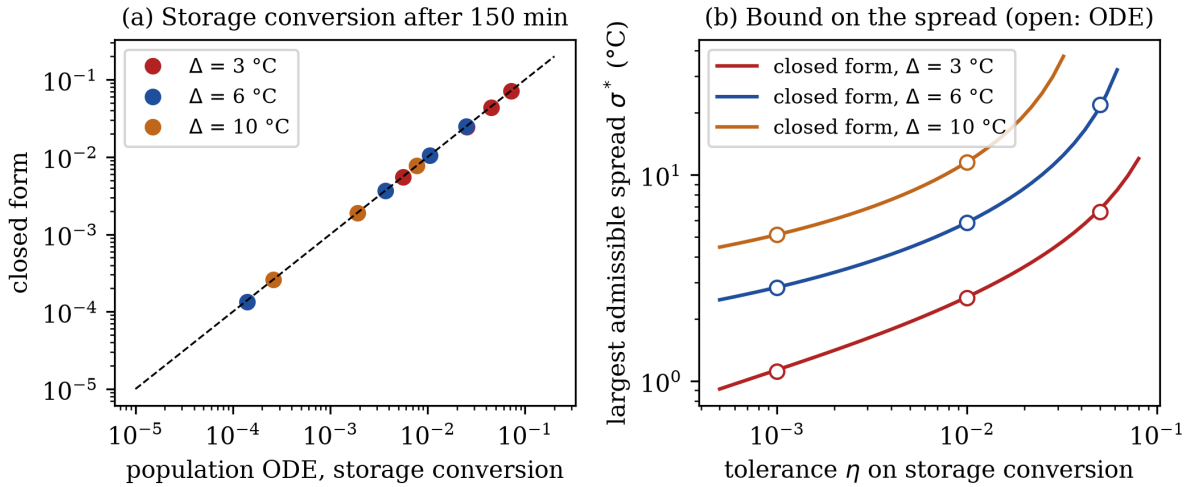


Fig. 2 (a) Closed-form storage conversion Eq. (11) against the population ODE for margins of 3, 6 and 10 °C. (b) Largest admissible spread σ^* (Eq. (12)) as a function of the tolerance η ; open symbols are roots of the population ODE.

D. Proposition 4: thermal-feedback criticality, overshoot, and Arrhenius correction

When the thermal time constant is short compared with the reaction time ($\Theta \ll 1$), the temperature rise $\theta = (T - T_\infty)/\Delta T_{ad}$ of the exponential comparison model follows the reaction quasi-statically with the

conversion frozen, $\theta = \Theta r(\alpha) \exp(\psi\theta)$, where $r(\alpha) = (\varepsilon + \alpha)(1 - \alpha)^n$ is the isothermal rate in units of k_2 . Multiplying by ψ gives $\psi\theta \exp(-\psi\theta) = \Pi r$, whose solution, if $\Pi r \leq 1/\varepsilon$, is

$$\psi\theta = -W_0(-\Pi r(\alpha)). \quad (13)$$

Proposition 4. For the ungated exponential model in the quasi-steady limit with frozen conversion, the low-temperature algebraic branch of Eq. (13) ceases to exist when $\Pi r(\alpha) = 1/\varepsilon$. Maximizing r over α gives the limiting fold criterion. For $r = (\varepsilon + \alpha)(1 - \alpha)^n$ the interior maximum is at $\alpha^* = (1 - n\varepsilon)/(1 + n)$, provided $n\varepsilon < 1$, with $r_{\max} = n^n(1 + \varepsilon)^{n+1}/(1 + n)^{n+1}$, so

$$\Pi_c = \frac{1}{\varepsilon r_{\max}} = \frac{(1 + n)^{1+n}}{e n^n (1 + \varepsilon)^{1+n}}, \quad \Pi_c \rightarrow e^{-1} \frac{(1 + n)^{1+n}}{n^n} \quad (\varepsilon \rightarrow 0). \quad (14)$$

The finite- ε value is 1.4627, 1.9641 and 2.4610 for $n = 1, 3/2$ and 2 at $\varepsilon = 3 \times 10^{-3}$, against the $\varepsilon \rightarrow 0$ limits 1.4715, 1.9789 and 2.4832; the finite- ε formula explains why the integrated values of Table 4 at small Θ lie slightly below the limits. Below Π_c the quasi-steady peak is $\psi\theta_{\text{peak}} = -W_0(-\Pi/(e\Pi_c))$, and close to the fold it follows $\psi\theta_{\text{peak}} \approx 1 - [2(1 - \Pi/\Pi_c)]^{1/2}$, the normal form of a saddle-node bifurcation of the algebraic equation [19, 20]. This is a statement about the frozen-conversion algebraic manifold. For a finite batch the reactant is consumed and the temperature rise is bounded by the available reaction heat, so whether the fold marks ignition must be tested by integration. We therefore define an *operational* threshold as the Π (or ΔT_{ad}) at which the integrated peak rise exceeds $0.3\Delta T_{\text{ad}}$, and the conclusions about overshoot below are numerical. For comparison, the classical Semenov criterion for a source without consumption, $r = 1$ in the same units, gives $1/\varepsilon = 0.37$; consumption and autocatalysis raise the frozen-conversion critical number by the factor $1/r_{\max} \approx 5.4$ for $n = 3/2$ (reactant consumption in the Semenov problem was treated in Ref. [9]).

Verification in the exponential model. We integrated the exponential-form equations with isothermal start and bisected the operational threshold ($\varepsilon = 3 \times 10^{-3}$; Table 4). As $\Theta \rightarrow 0$ the operational Π_c converges to Eq. (14) with finite ε : at $\Theta = 0.003$ the values are 1.471, 1.974 and 2.472 against 1.463, 1.964 and 2.461 (differences 0.4–0.6 %). The threshold definition matters little ($\Pi_c = 1.974, 1.974, 1.992$ for 0.15, 0.3 and 0.6 of ΔT_{ad}). The finite- Θ shift is upward ($\Pi_c = 2.33$ at $\Theta = 0.1$ for $n = 3/2$); it is computed, not given in closed form. At $\Theta = 0.002$ the integrated peak at 0.5–0.98 Π_c follows the Lambert-W formula and approaches the square-root form (Fig. 3b).

Table 4 Critical thermal-feedback number in the exponential (Frank-Kamenetskii) model: operational thresholds from integrations (peak rise $0.3\Delta T_{\text{ad}}$, $\varepsilon = 3 \times 10^{-3}$) against Eq. (14) with finite ε and its $\varepsilon \rightarrow 0$ limit.

n	Θ	Π_c , integrated (operational)	Π_c , Eq. (14) finite ε	Π_c , $\varepsilon \rightarrow 0$
1.0	0.1	1.826	1.463	1.472
1.0	0.03	1.555	1.463	1.472

n	Θ	Π_c , integrated (operational)	Π_c , Eq. (14) finite ε	Π_c , $\varepsilon \rightarrow 0$
1.0	0.01	1.491	1.463	1.472
1.0	0.003	1.471	1.463	1.472
1.5	0.1	2.333	1.964	1.979
1.5	0.03	2.063	1.964	1.979
1.5	0.01	1.997	1.964	1.979
1.5	0.003	1.974	1.964	1.979
2.0	0.1	2.857	2.461	2.483
2.0	0.03	2.573	2.461	2.483
2.0	0.01	2.498	2.461	2.483
2.0	0.003	2.472	2.461	2.483

Arrhenius kinetics. For a single Arrhenius source at a temperature rise $x = \Delta T/T_\infty$ the exponent is $Ar \cdot x/(1 + x)$, against $Ar \cdot x$ in the exponential model. With $y = \psi\theta$ the frozen-conversion balance is $y \exp[-y/(1 + y/Ar)] = \Pi_r$, and its fold is at y_f satisfying $y = (1 + y/Ar)^2$; the critical number is then

$$\Pi_c^{Arr} = \Pi_c^{FK} \cdot e^{y_f} \exp\left(-\frac{y_f}{1 + y_f/Ar}\right) \approx \Pi_c^{FK} \exp\left(\frac{1}{Ar}\right) \quad (15)$$

The factor $\exp(1/Ar)$ is only the first-order (large- Ar) estimate of the first factor; for $Ar = 18.0$ – 19.9 the two differ by about 0.3 %. With distinct activation energies of the competing rate terms the fold condition must be solved for the actual sum of rates. We tested the correction with the true Arrhenius model (ungated, isothermal start, both k_1 and k_2 Arrhenius) at 120, 140 and 160 °C ($Ar = 18.0$ – 19.9) and $\Theta = 0.01, 0.03, 0.1$, bisecting the adiabatic rise to the same operational threshold (Table 5). The prediction $\Pi_c^{FK}(\Theta)$ times the fold factor is within 5.0 % of the Arrhenius threshold in every one of the nine cases (times $\exp(1/Ar)$: within 5.4 %; the uncorrected exponential value is off by 6–11 %). This is empirical agreement over the tested grid; the correction has not been tested at small Ar or for other Θ . Release lag raised the operational threshold in the cases tested: at $\Theta = 0.03$ it increased Π_c from 2.174, 2.188 and 2.200 to 2.184, 2.199 and 2.212 at 120, 140 and 160 °C. No general inequality is claimed, because a smaller availability at the same conversion does not by itself show that the temperature histories compare.

Table 5 Operational critical number of the Arrhenius model (ungated, isothermal start; peak rise $0.3\Delta T_{ad}$) against the exponential-model value $FK(\Theta)$ multiplied by the frozen-fold factor of Eq. (15) and by $\exp(1/Ar)$ (difference = prediction/Arrhenius – 1), and the effect of a release lag.

T (°C)	Θ	Ar	Π_c Arrhenius	Π_c FK(Θ)	$\times$ fold factor	Difference	$\times \exp(1/Ar)$	Difference	Π_c , release lag
120	0.01	19.9	2.093	1.973	2.080	-0.6 %	2.074	-0.9 %	–
120	0.03	19.9	2.174	2.039	2.150	-1.1 %	2.144	-1.4 %	2.184
120	0.1	19.9	2.545	2.308	2.433	-4.4 %	2.427	-4.6 %	–
140	0.01	18.9	2.106	1.980	2.094	-0.6 %	2.088	-0.9 %	–
140	0.03	18.9	2.188	2.047	2.165	-1.1 %	2.158	-1.4 %	2.199
140	0.1	18.9	2.566	2.316	2.449	-4.6 %	2.441	-4.9 %	–
160	0.01	18.0	2.117	1.986	2.106	-0.5 %	2.099	-0.8 %	–
160	0.03	18.0	2.200	2.053	2.177	-1.0 %	2.169	-1.4 %	2.212
160	0.1	18.0	2.586	2.322	2.462	-4.8 %	2.454	-5.1 %	–

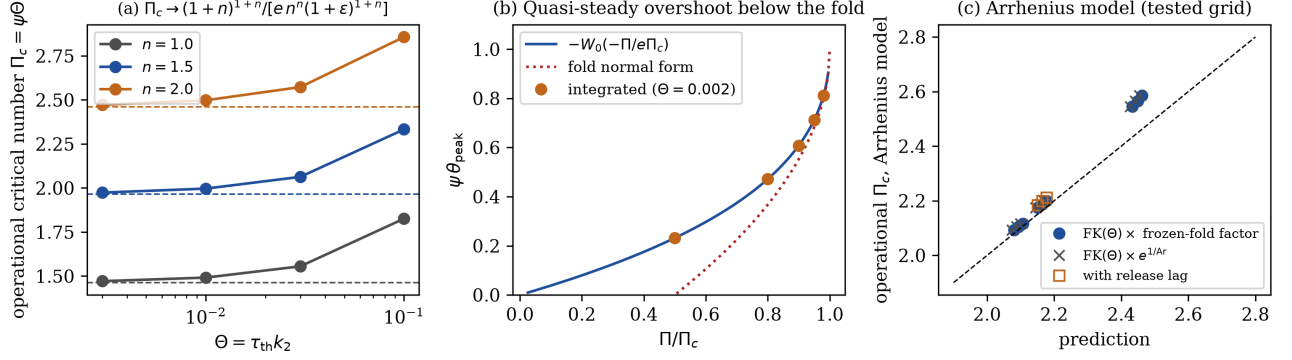


Fig. 3 (a) Operational critical number against Θ for $n = 1, 3/2$ and 2 (points: integrations; dashed: finite- ε Eq. (14)). (b) Quasi-steady overshoot below Π_c : Lambert-W formula, fold normal form, and integrations. (c) Arrhenius operational critical number against $FK(\Theta)$ times the frozen-fold factor.

E. Proposition 5: continuous-flow reactor, steady states, stability, and hysteresis

Let fresh resin of conversion 0 enter at the wall temperature with residence time τ_{res} , and let $\vartheta = \tau_{th}/\tau_{res}$ and $D = k_2\tau_{res}$. In units of τ_{res} and in the exponential model,

$$\frac{d\alpha}{dt'} = Dr - \alpha, \quad \frac{d\theta}{dt'} = Dr - \theta \left(1 + \frac{1}{\vartheta} \right), \quad r = (\varepsilon + \alpha)(1 - \alpha)^n e^{\psi\theta}. \quad (16)$$

Control parameter. D is swept with ϑ and ψ held fixed. Physically this is a sweep of the specimen temperature T_∞ at fixed τ_{th} and τ_{res} : D and Θ are both proportional to $k_2(T_\infty)$, so ϑ is constant, and in the exponential model a change of $\ln D$ by one unit corresponds to a temperature change of $RT^2/E_2 = 21.8$ K at 140 °C. A sweep of the flow rate at fixed temperature changes τ_{res} and therefore ϑ ; it is a different parameter path and is not analyzed here.

Proposition 5 (steady states). The steady states satisfy $\theta = \vartheta\alpha/(1 + \vartheta) = \kappa\alpha/\psi$ and are given parametrically by

$$D = \frac{\alpha}{(\varepsilon + \alpha)(1 - \alpha)^n \exp(\kappa\alpha)}, \quad \kappa = \frac{\psi\vartheta}{1 + \vartheta}. \quad (17)$$

Proof. At steady state the conversion balance gives $rD = \alpha$, and the heat balance gives $\theta(1 + 1/\vartheta) = rD = \alpha$, so $\theta = \vartheta\alpha/(1 + \vartheta) = \kappa\alpha/\psi$; substituting $\psi\theta = \kappa\alpha$ into $rD = \alpha$ gives the expression. ■ $D(\alpha)$ is non-monotonic, so three steady states coexist in a window of D , if and only if $d \ln D/d\alpha = 0$ has two roots, $n/(1 - \alpha) + \varepsilon/[\alpha(\varepsilon + \alpha)] = \kappa$. The left side has a single minimum, so three equilibria exist if and only if $\kappa > \kappa_c = \min_\alpha[n/(1 - \alpha) + \varepsilon/(\alpha(\varepsilon + \alpha))]$ ($\kappa_c = 1.894$ for $n = 3/2$ and $\varepsilon = 3 \times 10^{-3}$; $\kappa_c \rightarrow n$ as $\varepsilon \rightarrow 0$). Equilibrium multiplicity does not by itself establish two stable states or switching at both folds.

Stability. Let $q = \alpha/(\varepsilon + \alpha) - n\alpha/(1 - \alpha)$. Differentiating Eq. (16) at a steady state, with D and ϑ fixed, gives the Jacobian in the variables (α, θ)

$$J = \begin{bmatrix} q - 1 & \psi\alpha \\ q & \psi\alpha - 1 - 1/\vartheta \end{bmatrix} \quad (18)$$

with trace $q + \psi\alpha - 2 - 1/\vartheta$ and determinant $(1 + 1/\vartheta)(1 - q) - \psi\alpha$. At a fold the determinant vanishes and the other eigenvalue equals the trace. The lower fold has a negative trace for all cases below, so it is a saddle-node at which the lower (unreacted) state disappears. The upper fold has a negative trace, and is a saddle-node at which a stable upper state disappears, only for ψ below about 7.01 ($\vartheta = 1$); above it the trace at the upper fold is positive (+6.50 for $\psi = 20$: $\alpha = 0.8499$, $\ln D = -5.658$), so the upper branch near its fold is unstable and becomes stable only above a larger conversion α_H at which the trace vanishes ($\alpha_H = 0.9158$, $\ln D_H = -5.450$ for $\psi = 20$, with determinant $14.3 > 0$, a Hopf-type point). The window of two stable states is therefore $(\ln D_{\text{ext}}, \ln D_{\text{ign}})$, where $\ln D_{\text{ign}}$ is the lower (ignition) fold and $\ln D_{\text{ext}}$ is the upper fold when its trace is negative and $\ln D_H$ otherwise (Table 6).

Table 6 Steady states and stability of the continuous-flow reactor ($\vartheta = 1$, $n = 3/2$, $\varepsilon = 3 \times 10^{-3}$, $\kappa_c = 1.894$). The upper branch is stable from the upper fold ('fold') or from the Hopf-type point ('hopf'); the window width is the temperature interval $RT^2/E_2(\ln D_{\text{ign}} - \ln D_{\text{ext}})$ at 140 °C; the attractor loop area is $\int(\alpha_{\text{upper}} - \alpha_{\text{lower}}) d \ln D$ over the window of two stable states, and the fold-to-fold area uses the full multiplicity interval.

ψ	κ	Equilibria	Trace at upper fold	Upper stable from	In D window of two stable states	Window width (K)	Attractor loop area	Fold-to-fold area
3.0	1.50	1 ($\kappa < \kappa_c$)	—	—	—	—	0	0
4.5	2.25	3	-1.27	fold, $\alpha = 0.325$	-0.151 to -0.091	1.3	0.025	0.025
6.0	3.00	3	-0.51	fold, $\alpha = 0.498$	-0.466 to -0.130	7.3	0.227	0.227
8.0	4.00	3	+0.50	hopf, $\alpha = 0.710$	-0.987 to -0.169	17.9	0.678	0.710
12.0	6.00	3	+2.50	hopf, $\alpha = 0.844$	-2.280 to -0.225	44.9	1.921	2.040
20.0	10.00	3	+6.50	hopf, $\alpha = 0.916$	-5.450 to -0.307	112.3	5.047	5.236

Verification by attractors. We integrated Eq. (16) at fixed D from the unreacted state, from a mid-conversion state and from small perturbations of the upper steady state, for nine values of $\ln D$ around each window ($\psi = 4.5$ and 20). Two distinct long-time states were found for exactly those values of $\ln D$ that lie inside the window of Table 6 ($\psi = 4.5$: 3 of 3 inside values show two attractors; $\psi = 20$: 5 of 5), one state outside it, and no sustained oscillation anywhere (largest late-time oscillation amplitude in α : $2.8\text{e-}16$). At $\psi = 20$ the perturbed upper state collapses to the lower state for $\ln D$ just below $\ln D_H$, so the loss of stability at the Hopf-type point is, in these simulations, abrupt (subcritical-like), not a transition to a stable cycle; the unstable cycle, if any, was not computed.

Verification by sweeps. We swept $\ln D$ up and down at rates $v = 0.02, 0.01, 0.005, 0.002, 0.001, 0.0005$ and 0.0002 per residence time (protocol in Appendix A: start at the unreacted state, dwell 20 residence times at each turning point, sweep bounds ± 2 in $\ln D$ beyond the window). The loop area decreases monotonically with v but converges slowly. For $\psi = 20$ it falls from 5.427 to 5.144 against the attractor

loop area 5.047 of Table 6; a fit $A(v) = A_0 + cv^p$ to the seven rates gives $A_0 = 5.119$ with $p = 0.62$, which a two-point linear intercept would not capture. The downward switching of the sweeps occurs at $\ln D = -5.54$ ($v = 0.0002$), between the Hopf-type point (-5.45) and the fold (-5.66), because the loss of stability near a Hopf-type point is delayed at finite rate. For $\psi = 4.5$, just above κ_c , the attractor loop area is 0.025 and the integrated area at $v = 0.0002$ is 0.037 (fit $A_0 = 0.025$, $p = 0.63$); for $\psi = 8$ the attractor loop area is 0.678, the integrated area at $v = 0.0002$ is 0.722 and the fit gives 0.706; for $\psi = 3$ ($\kappa = 1.50 < \kappa_c$) there is no static loop and the area at $v = 0.0002$ is 0.002 (fit -0.001). A finite-rate loop therefore exists below the threshold as well, and only the convergence with v distinguishes a static loop from a rate-dependent one (Table 7).

Table 7 Integrated hysteresis-loop area $\oint \alpha d \ln D$ of the sweeps at seven sweep rates v (per residence time), with the attractor loop area, the fold-to-fold multiplicity area and a power-law fit $A(v) = A_0 + cv^p$ ($\vartheta = 1$, $n = 3/2$, $\varepsilon = 3 \times 10^{-3}$).

ψ	κ	Attractor loop area	Fold-to-fold area	$v = 0.02$	0.01	0.005	0.002	0.001	0.0005	0.0002	Fit A_0 (p)
3.0	1.50	0.000	0.000	0.125	0.069	0.037	0.016	0.008	0.004	0.002	-0.001 (0.86)
4.5	2.25	0.025	0.025	0.248	0.170	0.118	0.077	0.059	0.047	0.037	0.025 (0.63)
8.0	4.00	0.678	0.710	0.976	0.884	0.823	0.773	0.751	0.736	0.722	0.706 (0.60)
20.0	10.00	5.047	5.236	5.427	5.340	5.246	5.186	5.168	5.149	5.144	5.119 (0.62)

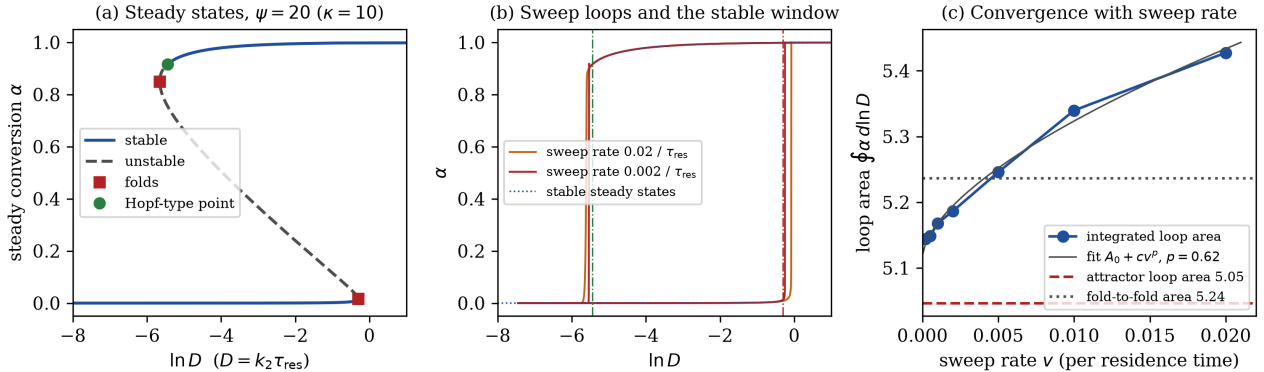


Fig. 4 (a) Steady-state curve of the flow reactor at $\psi = 20$ ($\kappa = 10$) with stable (solid) and unstable (dashed) segments, the two folds (squares) and the Hopf-type point (circle). (b) Sweep loops at two rates against the steady states. (c) Loop area against sweep rate, the power-law fit, and the attractor loop area.

F. Closed forms as fitting functions: a synthetic-data test

With a known logistic width w_0 , the storage curves depend on the capsule population only through the open fraction, i.e. through $z = (T_m - T_s)/\sigma_{\text{eff}}$ at each storage temperature, so the pair (T_m, σ) is determined only through the line $z(T_s) = (T_m - T_s)/\sigma_{\text{eff}}$; an unknown w_0 would add a third, confounded parameter. To see how well the pair can be recovered, synthetic isothermal storage curves were generated with the *population ODE* (not with the closed form) for $\sigma = 5.0$ °C, $T_m = 118$ °C and $w_0 = 0.5$ °C at three storage temperatures (112, 108, 105 °C) and five times (30–150 min), with Gaussian noise of 0.002 in

conversion, and fitted by least squares with the closed form (known w_0 ; 200 noise realizations). The estimates are $\hat{\sigma} = 7.5 \pm 2.6$ °C and $\hat{T}_m = 120.9 \pm 3.1$ °C with a correlation of 0.97 (Fig. 5): the two parameters are almost degenerate in this design, and $\hat{\sigma}$ is biased high along the degenerate direction. An independent constraint on T_m , here a prior of 0.5 °C (as from a calorimetric endotherm), markedly improved recovery in this tested design, to $\hat{\sigma} = 5.1 \pm 0.4$ °C. Other temperatures, times or noise levels may also improve identifiability; no claim is made about real calorimetry.

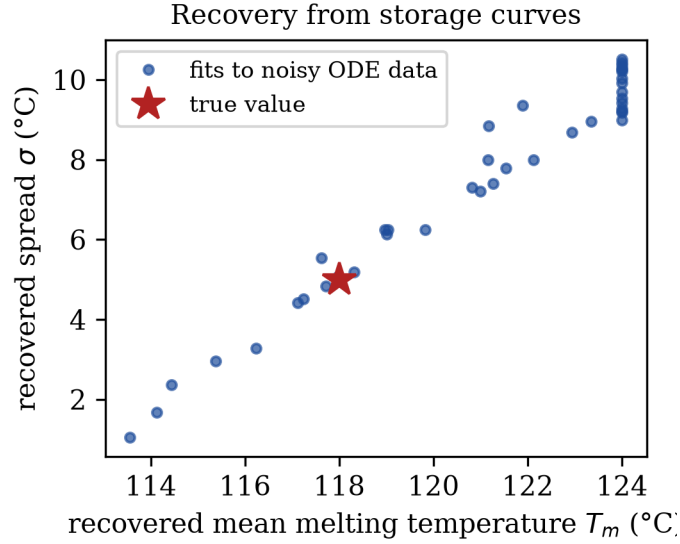


Fig. 5 Least-squares recovery of (T_m, σ) from noisy synthetic storage curves; each point is one noise realization (every fifth shown).

IV. Status of the claims and approximations

Table 8 Status of the principal results, their assumptions, and where they fail or have not been tested. All parameters are illustrative.

Result	Status	Assumption it rests on	Where it fails or is untested
Availability clock (Prop. 1)	Exact	Constant temperature; $a(0) = 0$, $C_i(0) = 0$; one availability function $\langle C \rangle(t)$	Temperature transients and exotherm; the delay is conversion dependent (Eq. (6)), and incomplete activation changes the clock
Isothermal time and sharpness (Prop. 2)	Exact (quadrature); $n = 1$ closed form; sharpness law asymptotic	$m = 1$; no diffusion control	Sharpness law only for $n = 1$ and $b \gg 9a$; vitrification not modelled
Storage conversion and σ^* (Prop. 3)	Linearized closed form; probit approximation; verified to a few percent	Gaussian spread; equal-mass independent capsules; small conversion	Large tolerance ($\eta \gtrsim 0.05$); skewed or bimodal distributions; capsule–capsule heat exchange
Critical number (Prop. 4)	Frozen-conversion fold, closed form; operational thresholds numerical	Exponential comparison model; ungated; $\Theta \rightarrow 0$	Finite Θ shifts Π_c upward (computed only); whether the fold marks ignition of a finite batch is numerical; spatial gradients absent
Arrhenius factor	Frozen-fold factor (exact for one Arrhenius source); $\exp(1/Ar)$ first-order	Single Arrhenius source for the fold	Tested only for $Ar = 18.0$ – 19.9 and $\Theta \leq 0.1$; not tested at small Ar
Flow reactor multiplicity (Prop. 5)	Exact for the stated model	Perfect mixing; exponential model; ungated; ϑ, ψ fixed	Real reactors: finite mixing time, wall–fluid temperature difference

Result	Status	Assumption it rests on	Where it fails or is untested
Flow reactor stability	Jacobian exact; stable window computed; checked by simulation	As above	Loss of stability at the Hopf-type point studied for $\psi = 20$ only; unstable cycle not computed
Parameter recovery (III F)	Numerical experiment	Gaussian noise; population ODE as truth; known w_0	Not a test on real calorimetry

Three approximations carry the paper: the lumped, well-mixed energy balance; the exponential linearization of the Arrhenius law near the fold; and the Gaussian, independent-capsule description of the population. The first is a deliberate scope limit and the second is quantified (Table 5); the third is tested only against population equations that use the same description.

V. Falsifiable signatures

Each proposition predicts a measurement that a standard calorimetric or rheological experiment can make; none has been made here. (1) *Availability clock*. Isothermal DSC at several temperatures above the melting endotherm, comparing the gated resin with the same resin with the catalyst dissolved, should give a delay that depends on conversion as Eq. (6) and approaches $\tau_{\text{rel}}(T)$ at late conversion; a conversion-dependent delay does not falsify the single-availability-function description, which predicts it. What would falsify it is a failure of the measured gated curves, replotted on the integrated-availability clock (measured independently, for example by release assays), to collapse onto the ungated curve. (2) *Storage bound*. The melting endotherm gives T_m and σ ; isothermal storage at a margin Δ for a time t should then give a conversion within a few tens of percent of Eq. (11); a systematically larger conversion would indicate a tail of capsules with low melting temperature that the Gaussian description misses. (3) *Critical thermal time constant*. At fixed T_∞ , specimens of increasing thickness (τ_{th} grows as thickness squared) should show a rapid rise of the peak temperature rise at an operational threshold near $\Pi_c/(\psi k_2)$, with the Arrhenius factor of Eq. (15); the abruptness of the rise, from $O(\Theta)$ to $O(\Delta T_{\text{ad}})$, is the signature of the fold. (4) *Static hysteresis*. In a continuously fed reactor with flow rate and wall time constant held fixed, a temperature sweep (the path of Section III E) should give a loop whose area converges to a nonzero limit as the sweep slows when ψ is sufficiently above its threshold, and to zero when $\kappa < \kappa_c$; a loop that closes as the sweep slows is rate dependent and not bistability. A sweep of the flow rate at fixed temperature is a different path (9 changes) and is not covered by the analysis.

VI. Open problems

(a) *Spatial extension*. The lumped balance replaces the classical slab, cylinder or sphere. The critical Frank-Kamenetskii parameter of a consuming, autocatalytic source in a spatially resolved specimen, and its relation to Π_c above, are open. (b) *Finite- Θ correction*. The upward shift of Π_c with Θ is computed numerically; a closed-form first correction would complete Proposition 4. (c) *Nature of the Hopf-type*

point. Whether the loss of stability of the upper branch of the flow reactor is sub- or supercritical in general, and the unstable cycle that mediates it, were studied only for $\psi = 20$. (d) *Non-Gaussian populations*. Proposition 3 needs only the lower tail of the melting-temperature distribution; skewed and bimodal distributions, and capsule–capsule heat exchange, are untested. (e) *Diffusion control and vitrification*. These stop the cure before completion and change the late-time tail of Proposition 2. (f) *Noise*. In small specimens or flow channels the fold may be crossed by fluctuations. (g) *Experiment*. The propositions are untested on a real encapsulated system; the signatures of Section V are the proposed tests.

VII. Conclusion

For the lumped model of a melt-gated autocatalytic cure, several statements can be made exactly or in closed form. The gated cure at constant temperature is the isothermal cure on the clock of integrated availability, and the delay of a fully open population is $\tau_{\text{rel}}[1 - \exp(-t/\tau_{\text{rel}})]$ (Proposition 1). The isothermal cure time is exact by quadrature, and the $n = 1$ sharpness is a logarithm of the rate-constant ratio (Proposition 2). The storage conversion of a Gaussian capsule population is $(a/b)[\exp(b f t_{\text{eff}}) - 1]$, which inverts into a bound on the melting-temperature spread (Proposition 3). The frozen-conversion thermal balance has a critical number $(1+n)^{1+n}/[e n^n(1+\epsilon)^{1+n}]$, with an Arrhenius correction that is close to $\exp(1/\text{Ar})$ in the tested grid (Proposition 4). The continuous-flow reactor has three equilibria above κ_c , but the upper branch can lose stability at a Hopf-type point before its fold, so the hysteresis window and its loop area must be computed from the stable states (Proposition 5). All results are tested against the full equations; none is tested against experiment, and the parameters are illustrative.

Declarations

Funding. The author received no funding for this work.

Competing interests. The author declares no competing interests.

Author contributions. Leon Sandler: conceptualization, methodology, software, formal analysis, investigation, writing – original draft, writing – review and editing, visualization.

Use of artificial intelligence. Claude Sonnet 5.5 (Anthropic) was used for derivations, code, figures and drafting as described in Section II C; the author is responsible for the content.

Data availability. The model code, the closed-form module, the tests, the figure scripts, the results file (results.json) and the manuscript builder are available at <https://github.com/sandlerleon/gated-autocatalysis-criticality> (release v1.1.0) and archived on Zenodo at

Appendix A. Numerical protocol, tolerances and convergence

Reference integrator. Eqs. (1)–(3) are integrated for all scenarios at once with a vectorized fixed-step fourth-order Runge–Kutta scheme (step 0.01 min for Sections III A and III B, 0.05 min for the population runs of III C; end time 150 min for storage, 300–600 min for cure times). Conversion is limited to $[0, 1]$ and the reaction heat is $\Delta T_{ad} d\alpha/dt$ without cut-off, so the energy balance is conserved: in the adiabatic limit the integrated temperature rise equals $\Delta T_{ad}\alpha$ to 6.2e-08 K (Section II, test_theory.py). Capsule populations use $K = 801$ deterministic Gaussian quantiles $z_i = \Phi^{-1}[(i - 1/2)/K]$. Convergence: relative to $K = 3201$ the storage conversion at $K = 801$ differs by at most 1.0e-02, and relative to step 0.0125 min the step 0.05 min differs by at most 3.1e-14 (Table 9; difference = value/reference – 1).

Adaptive integrations. The exponential-model criticality problem is integrated with LSODA (rtol 10^{-9} , atol 10^{-13} , max step 0.02, end time $40/k_2$); the Arrhenius criticality problem with LSODA (rtol 10^{-8} , atol 10^{-12} , end time $60/k_2$, temperature floored at 200 K in the rate); the flow reactor with LSODA (rtol 10^{-8} , atol 10^{-12} , max step $\min(0.5, 0.02/v)$); fixed-D attractor runs with LSODA (rtol 10^{-11} , atol 10^{-14} , end time 3000 residence times; the last 10 % is used to classify the state); constant-temperature delay checks with RK45 (rtol 10^{-12}).

Thresholds. The operational critical number is the Π (exponential model, bracket $[0.3, 8]$, 30 bisection steps) or the adiabatic rise (Arrhenius model, bracket $[3, 8000]$ K in logarithm, 26 steps) at which the integrated peak temperature rise $(T_{\max} - T_{\infty})/\Delta T_{ad}$ first exceeds 0.3, starting from $T(0) = T_{\infty}$. *Sweeps.* Start at $(\theta, \alpha) = (0, 0)$ at the lower bound of $\ln D$ (window lower edge minus 2), dwell 20 residence times, sweep up at rate v to the upper bound (window upper edge plus 2), dwell 20, sweep down, dwell 20; the area is the signed integral of $\alpha d \ln D$ over the two sweep legs (output step 0.05). *Design bound.* σ^* from Eq. (12) with $w_0 = 0.5$ °C; the ODE root is bracketed in $[0.3, 40]$ °C (xtol 10^{-3}). *Recovery.* 200 noise realizations (seed 20261007, prior test seed 20261008), bounded least squares. *Errors.* In every table, difference = closed form (or prediction)/numerical – 1. *Software.* Python 3.11.9, NumPy 2.4.6, SciPy 1.17.1; release v1.1.0 of the repository contains the exact scripts and results.json.

Table 9 Convergence of the reference integrator with population size and step (storage conversion after 150 min, $\Delta = 6$ °C, $w_0 = 0.5$ °C; the narrower spread $\sigma = 2$ °C is the tail-sensitive case).

Quantity	Setting	Value at setting	Reference	Relative difference
Storage conversion ($\sigma = 2$ °C, $\Delta = 6$ °C)	$K = 101$	0.000101	0.000142 ($K = 3201$)	-2.8e-01
Storage conversion ($\sigma = 2$ °C, $\Delta = 6$ °C)	$K = 201$	0.000124	0.000142 ($K = 3201$)	-1.3e-01
Storage conversion ($\sigma = 2$ °C, $\Delta = 6$ °C)	$K = 401$	0.000136	0.000142 ($K = 3201$)	-4.1e-02
Storage conversion ($\sigma = 2$ °C, $\Delta = 6$ °C)	$K = 801$	0.000140	0.000142 ($K = 3201$)	-1.0e-02
Storage conversion ($\sigma = 2$ °C, $\Delta = 6$ °C)	$K = 1601$	0.000142	0.000142 ($K = 3201$)	-2.0e-03

Quantity	Setting	Value at setting	Reference	Relative difference
Storage conversion ($\sigma = 2\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.2 min	0.000140	0.000140 (step 0.0125)	+6.7e-15
Storage conversion ($\sigma = 2\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.1 min	0.000140	0.000140 (step 0.0125)	-1.6e-14
Storage conversion ($\sigma = 2\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.05 min	0.000140	0.000140 (step 0.0125)	-3.1e-14
Storage conversion ($\sigma = 2\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.025 min	0.000140	0.000140 (step 0.0125)	-3.7e-14
Storage conversion ($\sigma = 2\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.0125 min	0.000140	0.000140 (step 0.0125)	+0.0e+00
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	K = 101	0.003681	0.003681 (K = 3201)	+1.0e-06
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	K = 201	0.003681	0.003681 (K = 3201)	+7.4e-08
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	K = 401	0.003681	0.003681 (K = 3201)	+5.3e-09
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	K = 801	0.003681	0.003681 (K = 3201)	+3.7e-10
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	K = 1601	0.003681	0.003681 (K = 3201)	+2.3e-11
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.2 min	0.003681	0.003681 (step 0.0125)	-8.2e-13
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.1 min	0.003681	0.003681 (step 0.0125)	-4.9e-14
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.05 min	0.003681	0.003681 (step 0.0125)	-2.2e-16
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.025 min	0.003681	0.003681 (step 0.0125)	+4.4e-16
Storage conversion ($\sigma = 4\text{ }^{\circ}\text{C}$, $\Delta = 6\text{ }^{\circ}\text{C}$)	step 0.0125 min	0.003681	0.003681 (step 0.0125)	+0.0e+00

Table 10 Run table of the numerical studies; every result is reproduced by code/run_all.py.

Study	Method	Grid	Compared with
Availability clock	RK4 (step 0.01 min) and RK45 (rtol 10^{-12}), constant T	T = 125, 140, 155 $^{\circ}\text{C}$; $\alpha = 0.001\text{--}0.9$	Eqs. (5), (6)
Isothermal cure	RK4, step 0.01 min	T = 110–155 $^{\circ}\text{C}$; n = 1, 3/2	Quadrature; Eqs. (7), (8), (9)
Storage conversion	RK4 population (K = 801), step 0.05 min	$\Delta = 3, 6, 10\text{ }^{\circ}\text{C}$; $\sigma = 2\text{--}10\text{ }^{\circ}\text{C}$; t = 150 min	Eq. (11)
Design bound	Root-finding on the population ODE	$\Delta = 3, 6, 10\text{ }^{\circ}\text{C}$; $\eta = 0.001, 0.01, 0.05$	Eq. (12)
Criticality (exponential model)	LSODA, bisection	n = 1, 3/2, 2; $\Theta = 0.1\text{--}0.003$	Eq. (14)
Criticality (Arrhenius)	LSODA, bisection	T = 120, 140, 160 $^{\circ}\text{C}$; $\Theta = 0.01\text{--}0.1$	Eq. (15)
Flow reactor	Jacobian; LSODA attractors and sweeps	$\psi = 3, 4.5, 6, 8, 12, 20$; $v = 0.02\text{--}0.001$	Eqs. (17), (18), attractor loop area
Parameter recovery	Bounded least squares, 200 noise draws	3 temperatures $\times$ 5 times	Truth

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